

Jaeseung Heo

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POSTECH (Pohang University of Science and Technology)

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EDUCATION

POSTECH

Integrated M.S./Ph.D. Program, Graduate School of Artificial Intelligence
Advisor: Prof. Dongwoo Kim

Pohang, South Korea

Mar, 2021 - Present

Expected Graduation: Feb, 2027

Sungkyunkwan University

B.S. in Computer Science and Engineering
B.S. in Systems Management Engineering (Double Major)

Seoul, South Korea

Mar, 2016 - Aug, 2020

RESEARCH EXPERIENCE

MATS Research

Exploration Phase, Neel Nanda Stream

Sep, 2026 - Present

RESEARCH INTERESTS

- **Training data attribution for LLMs:** Developing attribution methods applicable to LLMs, with the goal of scaling them to models with over 100B parameters.
- **The science of pre- and post-training:** Understanding which training data gives rise to which model behaviors, and whether filtering or perturbing the data can steer a model toward desired behaviors.
- **Mechanistic interpretability:** Explaining the mechanisms behind LLM behavior, with tools such as the Jacobian lens, sparse autoencoders, and cross-layer transcoders.

PUBLICATIONS

- **Scaling Influence Functions in LLMs through Eigenbasis-Corrected One-Bit Gradient Projection**
Jaeseung Heo, J Rosser, Dongwoo Kim
arXiv '26
- **Interaction-Aware Influence Functions for Group Attribution**
Jaeseung Heo, Kyeongheung Yun, Youngbin Choi, Sehyun Hwang, Jungseul Ok, Dongwoo Kim
ICML '26 MI Workshop
- **Posterior Label Smoothing for Node Classification**
Jaeseung Heo, MoonJeong Park, Dongwoo Kim
AAAI '26 (Oral)
- **Anatomy of Linearized Group Influence: From a Bregman Geometric Perspective**
Hyunho Choi, Jaeseung Heo, Youngbin Choi, MoonJeong Park, Dongwoo Kim
NeurIPS '26 ATTRIB Workshop
- **Long Live the Librarian! A Persistent Search Sub-Agent for Energy-Efficient Multi-Agent Software Engineering Systems**
Seunghyuk Cho, Sunghyun Choi, Jaeseung Heo, Youngbin Choi, Saemi Moon, MoonJeong Park, Dongwoo Kim
EMNLP '26 (Oral)
- **Transductive Generalization via Optimal Transport and Its Application to Graph Node Classification**
MoonJeong Park*, Seungbeom Lee*, Kyungmin Kim, Jaeseung Heo, Seunghyuk Cho, Shouheng Li, Sangdon Park, Dongwoo Kim
NeurIPS '26
- **Influence Functions for Edge Edits in Non-Convex Graph Neural Networks**
Jaeseung Heo, Kyeongheung Yun, Seokwon Yoon, MoonJeong Park, Jungseul Ok, Dongwoo Kim
NeurIPS '25

- **The Oversmoothing Fallacy: A Misguided Narrative in GNN Research**

MoonJeong Park, Sunghyun Choi, **Jaeseung Heo**, Eunhyeok Park, Dongwoo Kim

arXiv '25

- **EPIC: Graph Augmentation with Edit Path Interpolation via Learnable Cost**

Jaeseung Heo^{*}, Seungbeom Lee^{*}, Sungsoo Ahn, Dongwoo Kim

IJCAI '24

- **Mitigating Oversmoothing through Reverse Process of GNNs for Heterophilic Graphs**

MoonJeong Park, **Jaeseung Heo**, Dongwoo Kim

ICML '24

^{*}Equal contribution.

HONORS AND AWARDS

- **BK21 Outstanding Research Paper Award**, POSTECH

2025, 2026

- **Dean's List**, Sungkyunkwan University

2018, 2019, 2020

SERVICE AND TEACHING

- **Reviewer**, ICLR, ICML, ICML MI Workshop, NeurIPS, NeurIPS ATTRIB Workshop

- **Teaching Assistant**, CS273B: Convex Optimization, POSTECH

Fall 2022

- **Lecturer**, Youth AI & Big Data Academy, AI Expert Training Course, POSCO

2021, 2022

REFERENCES

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